

Geometry-resolved characterization of a dual-detector MA-XRF scanner

Dimitrije Pešić¹ Janko Vukobratović¹ Aleksandra Stojanović¹ Giulia Ristori²

Stefano Ridolfi³ Maja Gajić-Kvašček⁴ Goran Kvašček¹

¹School of Electrical Engineering, University of Belgrade ²Ars Mensurae, Rome

³IDArtScience Srl, Rome ⁴VINARH Center, Vinča Institute of Nuclear Sciences

One tilted scan separates detector from geometry

The two heads see the same spot from opposite take-off angles, so their count ratio $R = \text{upper head} / \text{lower head}$ mixes a detector part with a geometric part. Tilting the canvas moves only the geometric one: a second scan separates them, with no standards and no extra hardware.

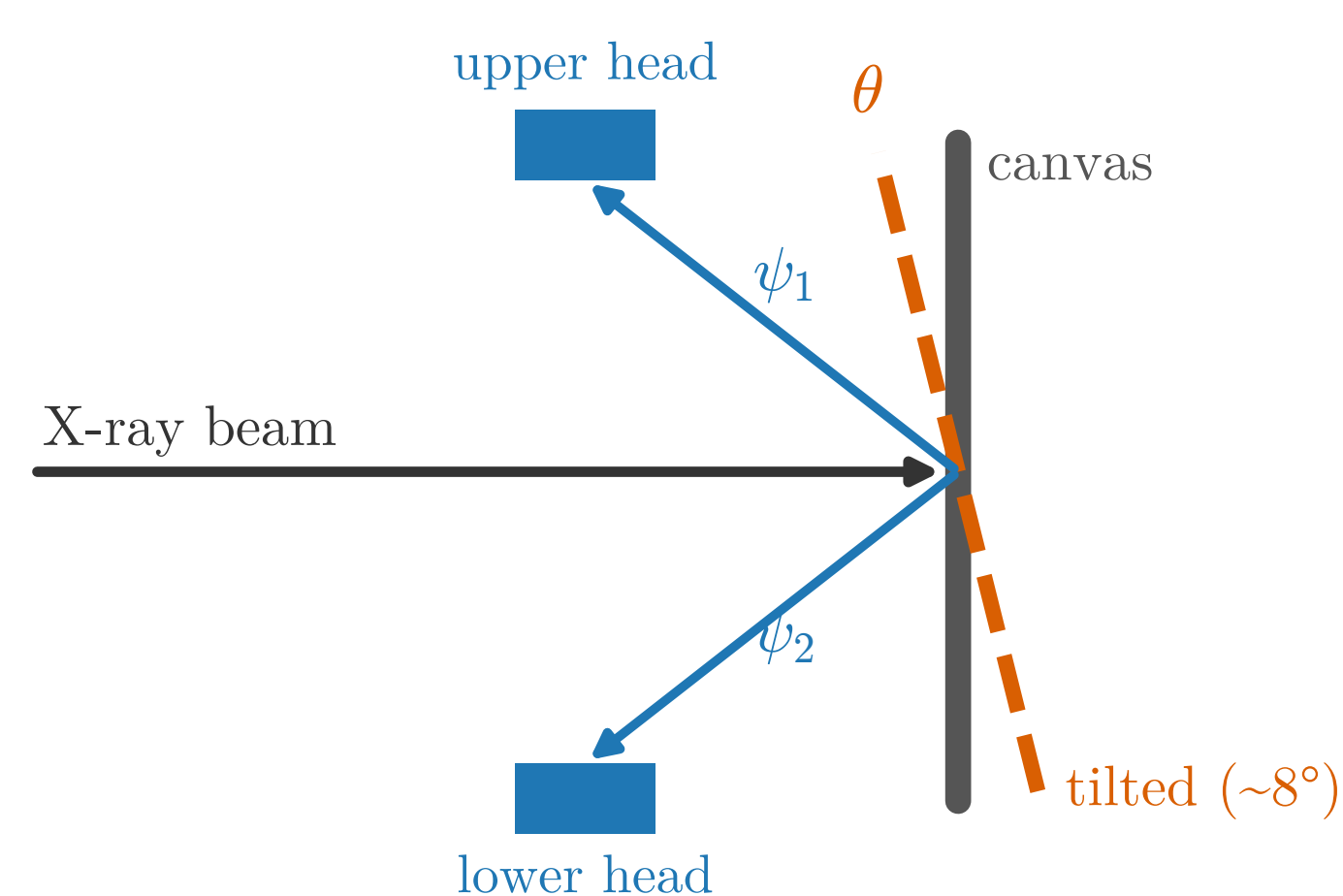


Figure 1: Tilting moves the geometric term only.

The two heads differ sixfold, and a physical model explains it

Stage 1 fits the frontal curve with a three-parameter detector model; stage 2 fits the tilt shift with a thick-sample model whose take-off angles move antisymmetrically with the tilt.

- R falls from **5.8** at Ca $K\alpha$ to **0.63** at Pb $L\gamma$: a sixfold different signal from the same pigment.
- Fitted absorber **$973 \pm 2 \mu\text{m}$** Be-equivalent, 80–120× any entrance window: extra air path, not a window.
- The tilt shift decays monotonically; $s = 0.53 \pm 0.10 \text{ deg/deg}$, and a model-free Gaussian-process fit agrees within 1.1σ .

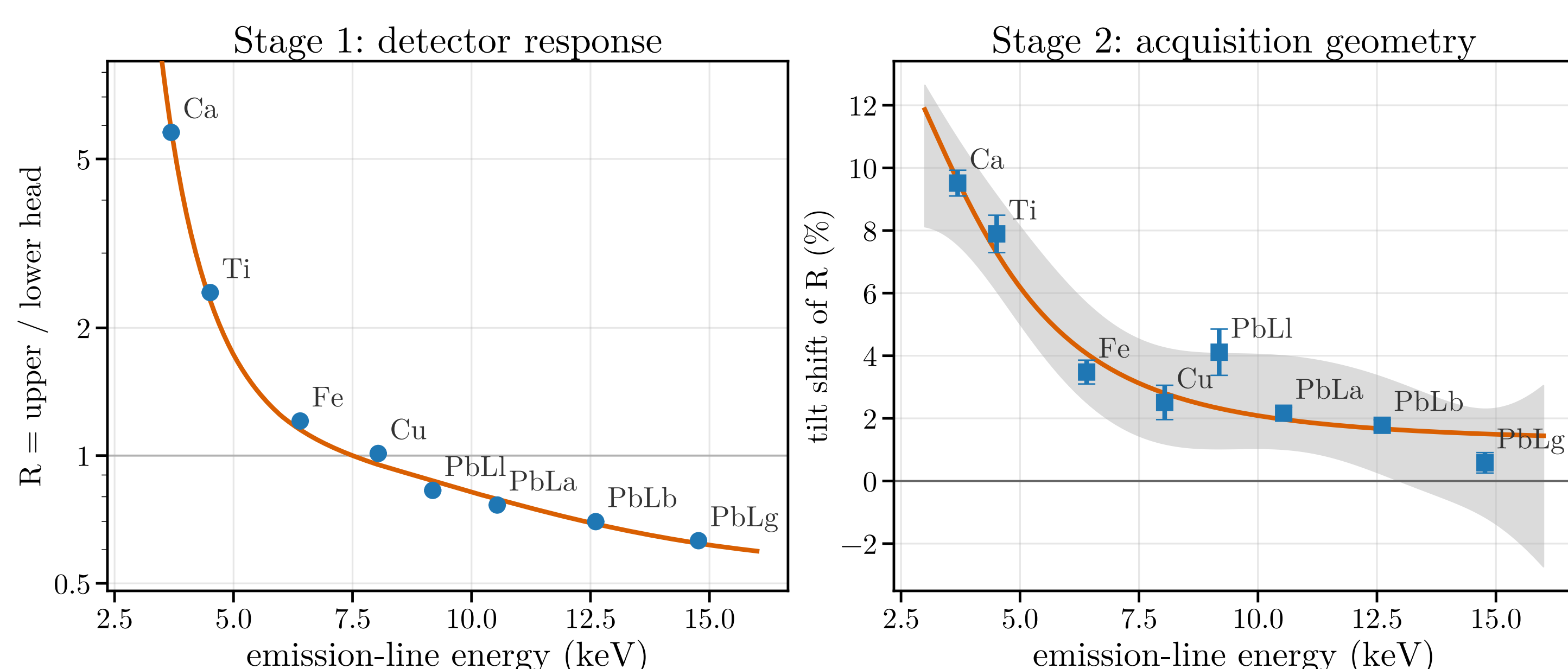


Figure 2: Points: measured on the registered overlap. Line: the physical model of each stage. Band: model-free GP fit, 2σ .

The discarded difference is also a relief map

Inverted through the measured tilt response, the per-pixel ratio residuals become repeated measurements of local surface slope: cross-scan $r = 0.73$, $\text{RMS} \approx 12^\circ$, $\chi^2/\text{dof} = 0.85$, and 88 % of pixels consistent with pure geometry: a relief map from one scan.

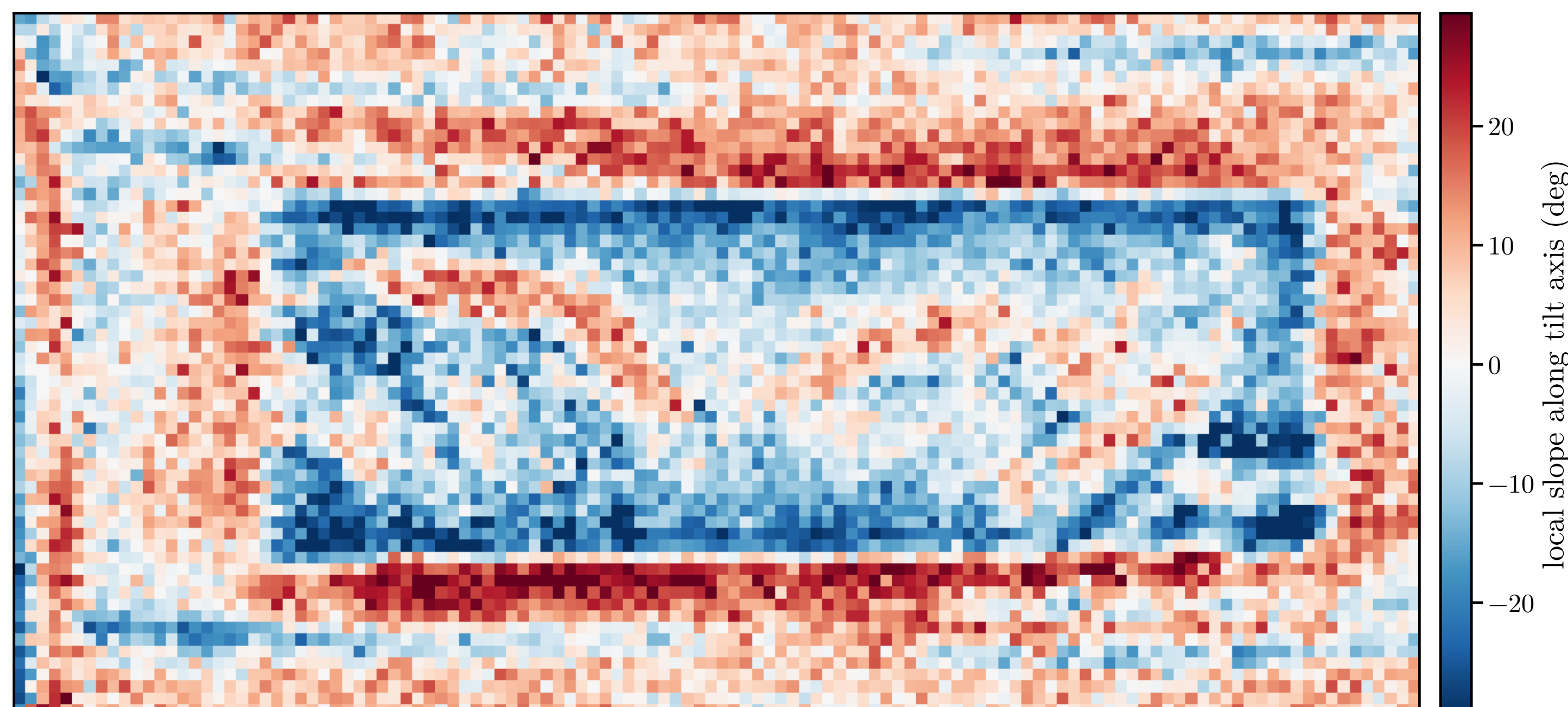


Figure 3: Local surface slope along the tilt axis, from one scan. Detector disagreement measures relief.

Two detectors, three scans, 7 200 pixels

Ars Mensurae MA-XRF scanner, two Amptek X-123SDD heads, 40 kV / 50 μA . Mock-up canvas, 60×120 pixels, 3.0 s dwell. Two frontal scans seven days apart set the repeatability baseline; a third is tilted forward by $\leq 8^\circ$. Ratios use the registered overlap (affine, NCC 0.965); full-frame ratios carry a field-of-view artifact that can flip the shift's sign.

Learned fusion beats summing by 10–18 % SNR

The two channels are conditionally independent Poisson views of the same spot, so Noise2Noise training needs no clean target. A 1D U-Net with loss weights from the measured $R(E)$ gains **+17.7 % mean SNR** over summing on held-out pixels; classical weighting gives +0.9 %.

- Six of eight lines gain: Pb $L\ell$ +69 %, Pb $L\gamma$ +33 %, Fe +28 %, Cu +17 %.
- An unweighted loss gives -0.1% : the gain is the measured variance structure, not the architecture. In acquisition time it is worth 39 % longer dwell.

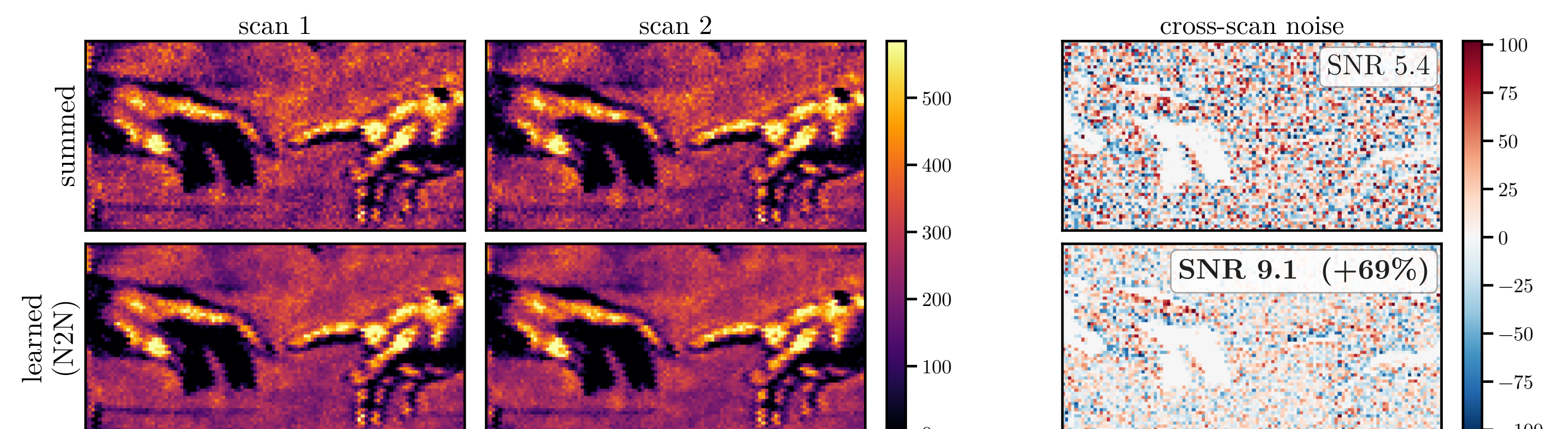


Figure 4: Pb $L\ell$ on pixels the network never saw: same signal, calmer noise. Maps in net counts; right, the cross-scan difference.

One degree of mounting error moves the maps

One degree of mounting error moves the element maps by up to **0.63 percentage points** (Ca +0.50 %/°, Ti +0.45 %/°, Pb $L\beta$ -0.13% /°), in the energy order the model predicts. These are lower bounds.

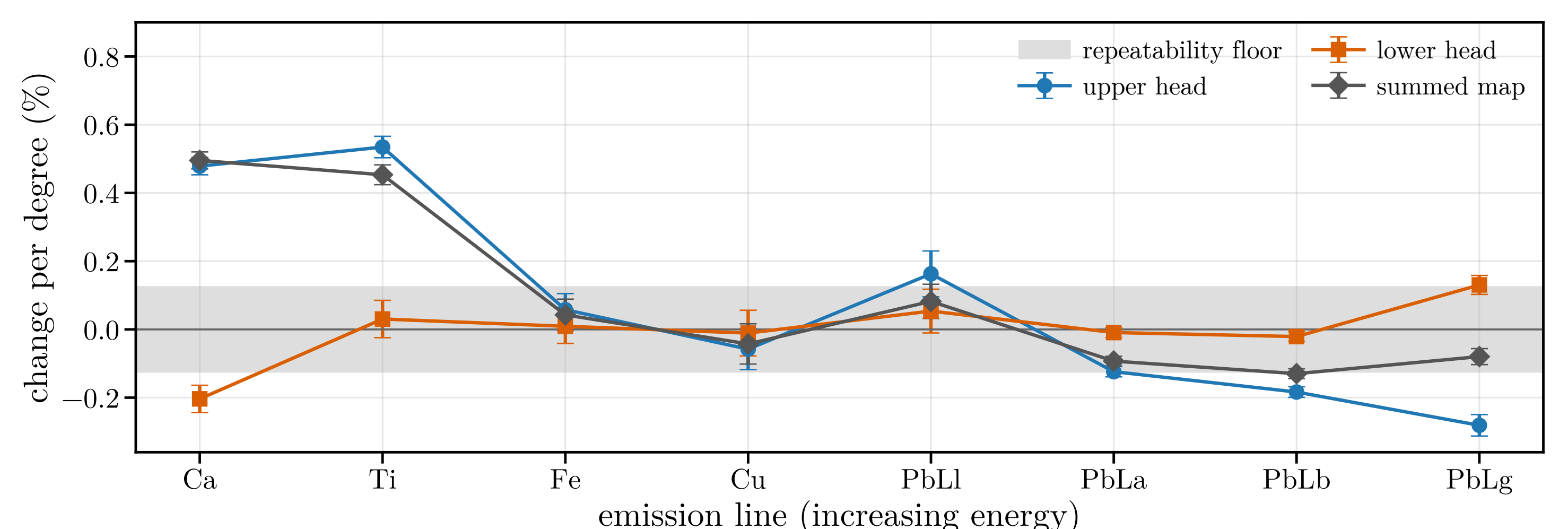


Figure 5: The light-element maps are the ones that move; the grey band is the repeatability floor.

The scanner characterizes itself from routine scans

The difference between a scanner's two detectors, normally summed away, plus one tilted scan characterizes the instrument for free: detector response, acquisition geometry, canvas topography, and a learned fusion that beats summing by 10–18 % SNR.

Field MA-XRF scanners can be characterized from their own routine scans, with mounting geometry in the error budget.

Acknowledgements. The research was partially conducted in the premises of the Palace of Science, Miodrag Kostić Endowment. We thank Ars Mensurae for instrument access and the scans.